



UDAY PRAKASH RASTOGI
B.Tech. - Computer Science & Engineering
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PROFESSIONAL SUMMARY

Motivated and detail-oriented Computer Science student with strong expertise in full-stack development and Artificial Intelligence, including hands-on experience in building real-world applications such as a RAG-based legal advisor chatbot and a peer-to-peer skill exchange platform. Skilled in designing scalable systems, developing RESTful APIs, and implementing intelligent solutions using modern technologies.

Passionate about solving real-world problems through technology, particularly in AI-driven systems and backend development. Seeking opportunities to contribute to innovative projects, enhance technical expertise, and deliver impactful solutions while growing within a dynamic and forward-thinking organization.

KEY EXPERTISE

Artificial Intelligence MERN Stack Node.js React.js Python MongoDB Problem Solving Langchain RAG
LLM SQL

EDUCATION

Oriental Group of Institutes B.Tech. - Computer Science & Engineering - Oriental Institute of Science and Technology, Bhopal CGPA: 7.64 / 10	2027
Karnal International School, Karnal 12 th CBSE Percentage: 72.40 / 100	2023
Shivalik Convent School Aliyaspur Ambala, Ambala 10 th CBSE Percentage: 78.60 / 100	2021

AWARDS AND SCHOLARSHIPS

- Academic project recognized and published in International Journal of Research Publication and Reviews (IJRPR)
- Secured Grade A (91.5%) for research work on AI-based legal chatbot

PROJECTS

Supply Chain Openenv Team Size: 3 Key Skills: JavaScript Python HTML Project Link: https://huggingface.co/spaces/udayprakashrastogi2005/supply-chain-openenv-final **Supply Chain OpenEnv – Simulation & Optimization Platform** Supply Chain OpenEnv is a system designed to model, simulate, and analyze supply chain operations in a controlled environment. The project focuses on representing real-world supply chain workflows such as inventory flow, demand-supply balancing, and process optimization through a structured and programmable environment. The application leverages modern development practices to handle data processing, environment simulation, and system interactions. It demonstrates the implementation of backend logic for managing supply chain states, along with frontend components for visualization and user interaction. The platform is structured to support experimentation with different supply chain strategies, enabling better decision-making and performance evaluation. This project highlights skills in system design, data modeling, and simulation-based problem solving, along with proficiency in building scalable applications. It reflects an understanding of how complex real-world processes like supply chains can be abstracted into programmable environments for analysis, optimization, and research purposes.	25 Mar, 2026 - 12 Apr, 2026
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Swapskill Team Size: 3 Key Skills: Typescript CSS Firebase Socket.io JWT MongoDB Mongoose Node.js Express.js Vercel Project Link: http://swapskill.vercel.app/	12 Jul, 2025 - 12 Jul, 2025
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****SkillSwap – Peer-to-Peer Skill Exchange Platform****

SkillSwap is a full-stack web application designed to enable users to exchange skills and knowledge through a collaborative, community-driven platform. The application allows users to create profiles, list skills they can offer, and discover other users with complementary learning needs, fostering a mutually beneficial learning environment without monetary transactions.

The platform is built using a modern MERN stack architecture, incorporating ****React.js**** for a dynamic and responsive frontend, and ****Node.js**** with ****Express.js**** for scalable backend services. ****MongoDB**** is utilized for efficient data storage and management, with ****Mongoose**** enabling structured schema modeling.

Key features include secure user authentication using JSON Web Tokens (JWT), real-time communication powered by Socket.io, and a structured system for managing skill listings, user interactions, and session scheduling. The application emphasizes seamless user experience, responsive design, and efficient API-driven communication between client and server.

SkillSwap demonstrates strong proficiency in full-stack development, RESTful API design, real-time system implementation, and modern UI/UX practices. It highlights the ability to design and deploy scalable web applications while addressing real-world use cases in collaborative learning and knowledge sharing.

PUBLICATIONS / RESEARCH / WHITE PAPERS

An Advanced AI-Powered Legal Advisor Chatbot for Rural India

12 Dec, 2025

Research Paper | International Journal of Research Publication and Reviews | **Mentor:** Prof. Neelesh Ray | **No. of Authors:** 3

Key Skills: NLP RAG LLM FAISS AI OCR Python FastAPI Langchain

This research addresses the critical issue of legal illiteracy and lack of access to legal services in rural India. Existing AI-based legal systems primarily target legal professionals and fail to provide accurate, accessible, and multilingual support for common citizens. Additionally, most systems suffer from hallucination issues, lack voice interaction, and do not ensure trust through verified sources.

To bridge this gap, we propose an advanced AI-powered legal advisor chatbot based on Retrieval-Augmented Generation (RAG). The system integrates DeepSeek R1 with FAISS-based vector search to retrieve relevant legal documents and generate accurate, context-aware responses with proper citations. It supports multilingual interaction, voice input/output, and document analysis using OCR, making it highly accessible for low-literacy users.

The proposed system demonstrates strong performance with 92% accuracy, 1.85-second response time, and significantly reduced operational cost. By combining natural language processing, information retrieval, and real-time AI capabilities, this solution provides a scalable and affordable approach to democratizing legal knowledge and empowering rural communities with reliable legal assistance.

ACHIEVEMENTS

- Developed an AI-powered legal advisor chatbot achieving 92% accuracy with multilingual and voice support
- Successfully built and deployed multiple full-stack applications using MERN stack
- Demonstrated strong problem-solving and system design skills through real-world projects

POSITIONS OF RESPONSIBILITY

Discipline Committee Member – TechFest

Ensured smooth execution of TechFest by managing crowd control, maintaining discipline, and coordinating with organizing teams. Played a key role in handling on-ground operations, resolving issues, and supporting event logistics for a seamless experience.

Team Leader – Hackathon Projects

Led a team in hackathon competitions, overseeing project planning, task distribution, and development. Coordinated team efforts under tight deadlines, contributed to technical implementation, and ensured successful project delivery through effective collaboration and problem-solving.

Google Ambassador

Represented Google initiatives at the campus level by promoting developer programs, organizing tech-related activities, and engaging students in learning opportunities. Helped build awareness around Google technologies and fostered a tech-driven community.

GirlScript Summer of Code Contributor

Contributed to open-source projects under GirlScript Summer of Code by collaborating with developers, resolving issues, and enhancing project features. Gained hands-on experience in Git, GitHub workflows, and real-world software development practices.

CO-CURRICULAR ACTIVITIES

- Actively involved in learning emerging technologies like AI, LLMs, and system design
- Active member of the discipline committee during college TechFest, responsible for event coordination, crowd management, and maintaining smooth execution of activities

EXTRA CURRICULAR ACTIVITIES

- Actively participated in dance performances during college cultural events, showcasing creativity, teamwork, and confidence
- Engaged in college-level cultural programs and performance

PERSONAL INTERESTS / HOBBIES

- Exploring new technologies (AI, startups, tech trends)

WEB LINKS

- Github - <https://github.com/udayrastogi0531>

PERSONAL DETAILS

Gender: Male	Date of Birth: 05 Feb, 2005
Marital Status: Single	Permanent Address: Madhav JI Prakash Store Sakri Kudra,
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